MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **WEEKLY CHAPTER TEST – 03 (POLYNOMIALS)** | **Time:** 45 Minutes

Class: Class X (CBSE) | **Chapter 2: Polynomials** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Question 10(iii).
2. **Section A** comprises 5 questions (4 MCQs and 1 Assertion-Reason) of 1 mark each.
3. **Section B** comprises 1 VSAQ of 2 marks and 3 SA questions of 3 marks each.
4. **Section C** comprises 1 Applied Case Study with 3 sub-parts of 4 marks total.

SECTION A: Multiple Choice Questions (5 × 1 = 5 Marks)

Q1. If one zero of the polynomial $f(x) = (k^2+4)x^2 + 13x + 4k$ is reciprocal of the other, then $k =$: [1]

(A) 2 (B) -2 (C) 1 (D) -1

Q2. If α and β are the zeroes of the polynomial $f(x) = x^2 + x - 1$, then $\frac{1}{\alpha} + \frac{1}{\beta}$ is equal to: [1]

(A) 1 (B) -1 (C) 0 (D) 2

Q3. If one zero of the polynomial $f(x) = 5x^2 + 13x + k$ is reciprocal of the other, then the value of k is: [1]

(A) 0 (B) 5 (C) $\frac{1}{6}$ (D) 6

Q4. The graph of $y = p(x)$ intersects the x-axis at $(-3, 0)$ and $(5, 0)$. The number of zeroes of $p(x)$ is: [1]

(A) 0 (B) 1 (C) 2 (D) 3

Q5. **Assertion (A):** If the zeroes of $ax^2 + bx + c$ are equal in magnitude but opposite in sign, then $b = 0$.
Reason (R): Sum of zeroes is $-\frac{b}{a}$, and $\alpha + (-\alpha) = 0 \implies -\frac{b}{a} = 0 \implies b = 0$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Short Answer Type Questions (1 × 2 + 3 × 3 = 11 Marks)

Q6. If α, β are the zeroes of the polynomial $2y^2 + 7y + 5$, write the value of $\alpha + \beta + \alpha\beta$. [2]

Q7. Find the zeroes of the quadratic polynomial $f(x) = \sqrt{3}x^2 - 8x + 4\sqrt{3}$ and verify the relationship between the zeroes and the coefficients. [3]

Q8. If α and β are the zeroes of the quadratic polynomial $f(x) = x^2 - px + q$, prove that: [3]

$$\frac{\alpha^2}{\beta^2} + \frac{\beta^2}{\alpha^2} = \frac{p^4}{q^2} - \frac{4p^2}{q} + 2$$

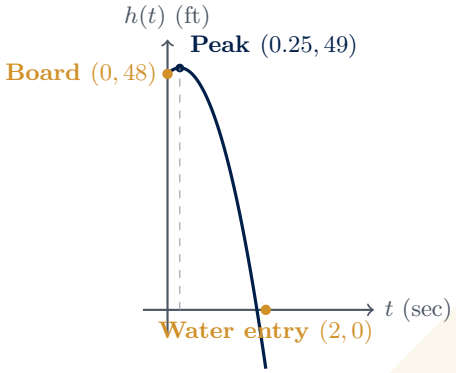
Q9. If α and β are the zeroes of the polynomial $x^2 - 6x + k$ such that $3\alpha + 2\beta = 20$, then find the value of k. [3]

SECTION C: Applied Case Study Analysis (1 × 4 = 4 Marks)

Question 10: Case Study Analysis — High-Diving Athlete Annie

At an Olympic aquatic centre, Annie dives from a high diving platform into a swimming pool. Her height h (in feet) above the water surface t seconds after leaving the diving board is modelled by the quadratic polynomial:

$$h(t) = -16t^2 + 8t + 48$$



Based on the above mathematical model, answer the following questions:

(i) What was Annie's initial height above the water level at the instant she jumped ($t = 0$)? [1]

(ii) At what time t will Annie hit the surface of the water in the pool? [1]

(iii) **(a)** A second diver, Rita, dives such that her height is modelled by a polynomial $p(t)$ whose zeroes are -1 and 2, and initial height is 32 ft. Find the polynomial $p(t)$. [2]

— OR —

(b) Determine the maximum clearance height above the water level achieved by Annie during her dive. [2]

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